

The Multiplicative Dissipation Cellular Automaton

A specification of the model as implemented

Abstract

A stochastic cellular automaton on a square lattice in which occupied cells belong to a two-level taxonomy of *species* and *sub-species*. Cells compete pairwise for territory, and the outcome of a contest is biased by a species-level energy variable α . Energy is produced in proportion to a species' area and dissipated in proportion to its internal diversity, so that a species which accumulates too many sub-species exhausts its energy and *fragments*: its sub-species are promoted, all at once, to independent species. Speciation is therefore not an elementary move but a catastrophe that follows a slow build-up of internal multiplicity — the “multiplicative dissipation” of the title. Three further mechanisms shape the texture of the result: neighbour support, which gives domains surface tension; a rare-type advantage within a species, which keeps its sub-species comparable in size and so makes fragmentation a real division; and a colour thermostat, which makes hue dynamical and heritable. Five of the model's rules are not fixed but switchable, and each alternative is set out here beside the default.

1 Overview

The lattice starts with a single occupied cell of a single species at its centre. Occupied cells spread into empty ground and contest each other's territory. Two mechanisms generate novelty. *Mutation* converts one cell into a new sub-species of its own species, adding to that species' internal diversity. *Fragmentation* converts an entire species' worth of accumulated diversity into new species at a stroke, when the species runs out of energy. Between these two mechanisms the single initial species can, in principle, give rise to indefinitely many; in practice it gives rise to tens of thousands in a few hundred sweeps.

Three further mechanisms shape what happens between those events. *Neighbour support* rewards a cell for being surrounded by its own kind, which gives domains surface tension and keeps them compact. An *occupancy bias* within a species favours whichever sub-species is locally rarer, which keeps internal diversity alive long enough to matter. And a *colour thermostat* governs how the hues of new species separate after a burst, so that the picture on screen reports the genealogy rather than merely labelling it.

2 State

2.1 Lattice

A square lattice of dimension $N \times N$, comprising N^2 cells, with N a power of two drawn from $\{64, 128, 256, 512\}$; the default is $N = 256$. The neighbourhood is **Moore** — the eight cells that touch, edges and corners alike — and the boundary is **periodic**: the lattice is a torus, with no edge and no corner. Each cell is either **empty** or **occupied**.

2.2 Taxonomy

Every occupied cell carries two labels: a **species** and, within that species, a **sub-species**. The two levels are nested — a mutation changes a cell's sub-species label while leaving its species

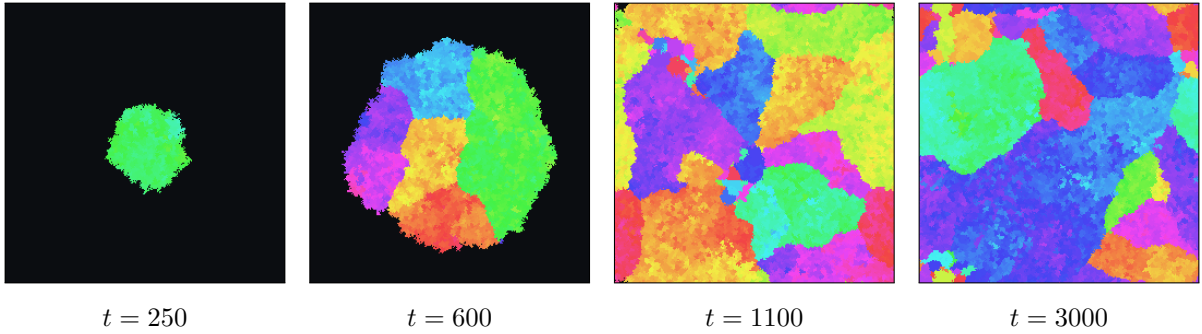


Figure 1: A run at the default settings on a 256^2 lattice. At $t = 250$ a single species holds six per cent of the ground and already carries 443 sub-species, visible as mottling within one narrow band of hue. By $t = 600$ it has burst six times and the fragments are still spreading into open ground — at this colonisation rate a species reaches the diversity that destroys it long before it reaches the walls. The lattice closes at about $t = 1100$; by $t = 3000$ the run holds some fifty living species out of 125,000 ever created.

label intact — and they differ only in how they interact (Section 5). Sub-species spread through the domain of their parent species by much the same mechanism by which the species spreads into empty ground.

In the lattice itself only the sub-species label is stored; the species is recovered from it, since each sub-species belongs to exactly one species at a time. When a species fragments, its sub-species are re-pointed at their new parents, and every cell follows without being touched.

2.3 Species-level bookkeeping

For each extant species i the model tracks

α_i	<i>energy</i> of species i : a real variable, evolving in time
N_i	number of lattice cells occupied by species i
X_i	number of <i>extant</i> sub-species of species i
φ_i	the species’ hue, in degrees on the colour circle

and, for display only, a serial **tag** that is never reused, and a count of the sub-species it has ever carried. Each sub-species in turn knows its own hue and the number of cells it occupies.

Note that $X_i \leq N_i$, with equality when every cell of the species is its own sub-species. The variables N_i and X_i are read off the lattice; α_i obeys its own equation of motion (Section 7).

X_i counts what is alive. There are two things “number of sub-species” could mean: those currently extant, or every one the species has ever produced. X_i is the extant count, and the difference is not cosmetic. A sub-species driven to extinction by internal competition stops costing its parent anything, so the coarsening of Section 5 is a genuine regulator of dissipation, and a species can hold its diversity — and hence its energy — in balance. Were the cumulative count used instead, the cost could only ever rise, and fragmentation would become certain from the moment a species began to diversify. The cumulative count is tracked, but only to be displayed.

2.4 Initial condition

One occupied cell at the centre of the grid, belonging to a single species with a single sub-species; all other cells empty. The founder’s hue is drawn uniformly from the circle, and its energy is set to α_0 .

3 Time

The time increment associated with one elementary competition event is

$$\Delta t = \frac{1}{N^2}, \quad (1)$$

so that one unit of model time — one **sweep** — corresponds to N^2 elementary events, i.e. on average one attempt per lattice site.

Competition is a discrete sequence of events; mutation, energy and colour are continuous processes running alongside it. Rather than interleave them event by event, the model advances in **chunks**. A chunk resolves K elementary contests, then advances the clock by $K\Delta t$ and applies the mutation, energy and colour updates once each over that whole interval. The chunk size is

$$K = \min(\max(256, 4S), 4096), \quad (2)$$

where S is the current number of living species: the passes that cost $O(S)$ are run less often when there are more species to pay for. The splitting is the only approximation in the model's time-stepping, and its error is bounded by the amount α can move within one chunk — at the default settings, a few parts in a thousand.

4 Parameters

Symbol	Meaning	Default	Range
N	linear grid dimension	256	64–512
d	resistance of empty ground; $P_{\text{expand}} = 1 - d$	0.90	0–1
E	energy supplied per occupied cell per sweep	0.0316	10^{-3} – $10^{-0.5}$
β	cost per sub-species per sweep	0.288	10^{-1} – $10^{1.5}$
m	mutation-rate constant	8.9×10^{-5}	10^{-5} – 10^{-1}
δ	fragmentation-rate constant	7.1×10^{-3}	10^{-3} – 10^1
α_0	seed and reset energy	0.01	10^{-2} – 10^2
λ	rare-sub-species advantage	1.14	0–2
θ	neighbour-support exponent	3	0–3
<i>Colour, thermostat rule</i>			
D	burst hue dispersal, degrees	90	0–360
T	colour settle time, sweeps	2	0.32–200
a	settle pull (labelled α in the panel)	2	0–12
k	number of settled hues	12	2–36
<i>Colour, other rules</i>			
g	burst hue spread	2	0–5
σ	burst hue jitter, degrees	47	0–180

All rates are per unit of model time, i.e. per sweep. The sliders for E , β , m , δ , α_0 and T travel on a logarithmic scale, since each matters across decades rather than across an interval.

5 Competition: the elementary update

At each iteration a cell is chosen uniformly at random from all N^2 , and one of its eight neighbours is chosen uniformly at random. If the two carry the same label, or both are empty, nothing happens. Otherwise the two cells **compete**, and the label of the winner is copied into the cell of the loser. Which of the two wins is probabilistic. There are three cases.

Type I: empty–occupied. The occupied cell takes the empty one with a fixed probability, independent of everything else. Empty ground behaves as a rival of fixed strength d — its *resistance* — so that

$$P_{\text{expand}} = 1 - d, \quad P_{\text{no expand}} = d. \quad (3)$$

At the default $d = 0.90$ a contested empty cell is taken one time in ten, which makes colonisation slow: the lattice takes some eleven hundred sweeps to fill. Colonisation carries no dependence on energy. This is a deliberate restriction: α governs conflict between species and nothing else, so growth into open ground proceeds at a uniform rate and a species cannot widen its frontier by being rich. It can only take ground that is already held.

The asymmetry of this case must be stated explicitly: if the occupied cell loses, the outcome is simply that *no expansion occurs*. The occupied cell is not replaced by an empty one; empty ground never invades. Note also that either of the two cells may be the focal one — an empty cell chosen first is colonised by an occupied neighbour just as readily — so the rule is symmetric in the pair even though its outcome is not.

Type II: two sub-species of the same species. Two sub-species of a common species share their parent’s energy, so α cannot separate them. Stripped of everything else the contest is therefore a fair coin,

$$P_{u \text{ wins}} = P_{v \text{ wins}} = \frac{1}{2}, \quad (4)$$

under which competition within a species is unbiased and sub-species boundaries perform a neutral, voter-model coarsening inside the species’ domain.

Two factors may be layered on top of that coin, each with its own control: how well each cell is supported by its immediate neighbours, and how rare each sub-species is within the parent. With both in force — the default — and writing u and v for the two sub-species, n_u and n_v for the number of Moore neighbours matching each at its own cell, and $|u|$, $|v|$ for the number of cells each occupies,

$$P_{u \text{ wins}} = \frac{(n_u + 1)^\theta e^{-\lambda|u|/\kappa_i}}{(n_u + 1)^\theta e^{-\lambda|u|/\kappa_i} + (n_v + 1)^\theta e^{-\lambda|v|/\kappa_i}}, \quad \kappa_i = \frac{N_i}{X_i}. \quad (5)$$

The parent’s energy has cancelled from numerator and denominator, as it must. The two factors are independent and are switched separately: the *neighbour support* control removes the first, which is equivalent to setting $\theta = 0$, and the *within-species* control’s *tie* setting removes the second, which is equivalent to $\lambda = 0$. Only when both are removed is (4) recovered exactly — support alone still favours the better-embedded cell, even under the tie setting.

The exponential factor is a **rare-type advantage**. Because $|u|$ appears with a negative sign, the *larger* sub-species is the one penalised: at the default $\lambda = 1.14$, a sub-species that exceeds its rival by one mean sub-species area wins only about a quarter of their contests. The normalisation κ_i is the mean sub-species area within the parent, so the bias reads differences in size relative to the species’ own internal scale rather than in absolute cells.

What the advantage changes is not how many sub-species a species carries but how large any one of them is allowed to grow. Under the tie setting, coarsening within a species does what a voter model does: one sub-species engulfs the domain, and most of the rest are recent mutations that have not yet been swallowed. At $t = 2000$ and the default parameters a single species has then taken the whole lattice, 88 per cent of it is one sub-species, and 96 per cent of its sub-species are lone cells. Under the advantage the largest species is instead a mosaic of comparable patches — median area ten cells, largest twenty-nine — because nothing can grow large without being penalised for it. Section 11 draws out why this matters: a burst divides a species along its sub-species, so it is their size distribution, and not their number, that decides how much of the parent survives the catastrophe.

Whichever settings are in force, this is the sole channel by which a sub-species can be driven extinct without the whole species being involved.

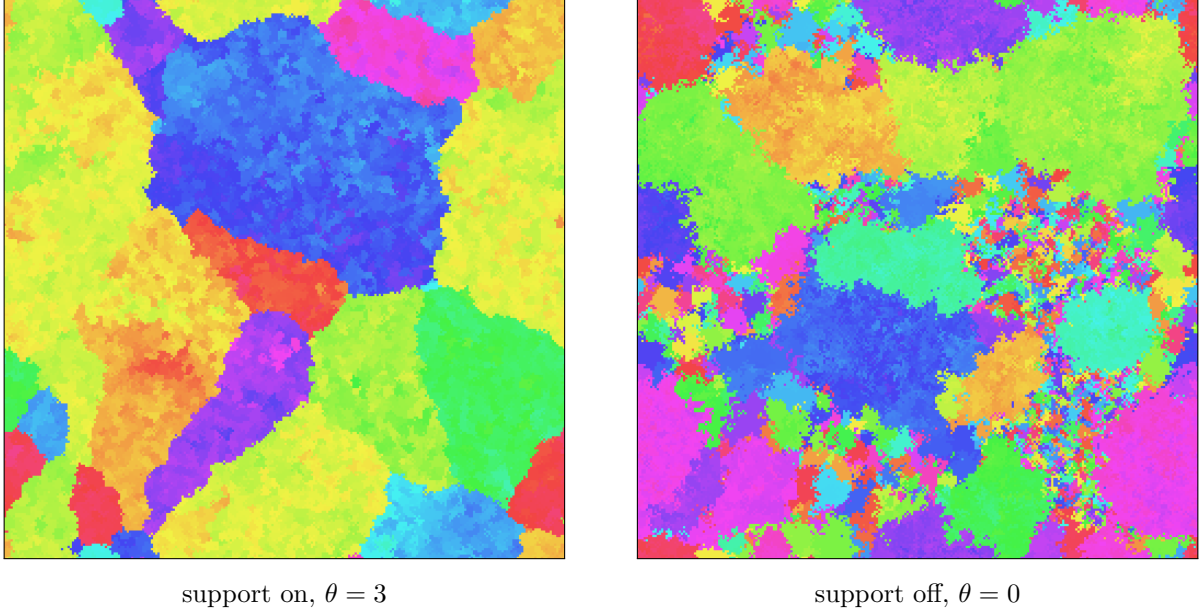


Figure 2: The same seed and the same time ($t = 2000$), with neighbour support on and off. With it, domains are compact and their boundaries short: seven per cent of occupied cells lie on a species boundary. Without it, boundaries wander, domains interpenetrate, and the boundary fraction rises to thirty-six per cent. The ragged state also carries far more species at once — 1335 against 26 — because nothing gathers a domain together and small holdings are never squeezed out.

Type III: two distinct species. For species A and B ,

$$P_{A \text{ wins}} = \frac{\alpha_A(n_A + 1)^\theta}{\alpha_A(n_A + 1)^\theta + \alpha_B(n_B + 1)^\theta}, \quad (6)$$

where n_A and n_B now count Moore neighbours of the *same species* at each of the two cells. A species with larger α is more “powerful”: it conquers its neighbours with greater likelihood. This is the sense in which α behaves like an energy. Should both strengths vanish, the contest is decided by a fair coin.

The occupancy bias does *not* reach this case. Rarity is an advantage inside a species, where it preserves the diversity that drives the model; between species it would merely blunt the effect of α , which is the quantity the model is about.

5.1 Neighbour support

Both Types II and III carry the factor $(n + 1)^\theta$, where $n \in \{0, \dots, 8\}$ counts the neighbours of the contesting cell that match it at the level of that contest — sub-species for Type II, species for Type III. Empty neighbours are not counted, so a cell on the frontier is weaker than one in the interior. The support factor therefore ranges over $[1, 9^\theta]$, and at the default $\theta = 3$ it spans nearly three orders of magnitude: a cell with all eight neighbours of its own kind beats an isolated intruder in better than 998 contests out of 1000.

The effect is surface tension. Without it, boundaries wander freely and domains dissolve into filigree; with it, a domain pays a price along its perimeter and none in its interior, so it rounds up, holds its shape, and can only be eaten from the edge. Setting $\theta = 0$ switches it off, and Types II and III revert to their unadorned forms.

5.2 A unified reading of the three rules

Types II and III are one rule. Give every occupied cell c , carrying sub-species u of species i , a strength

$$w(c) = \underbrace{\alpha_i}_{\text{species energy}} \times \underbrace{(n_c + 1)^\theta}_{\text{local support}} \times \underbrace{e^{-\lambda|u|/\kappa_i}}_{\text{rare-type discount}}, \quad (7)$$

and let the contest go to c with probability $w(c)/(w(c) + w(c'))$. Then:

Opponent	What survives in w	Recovers
a sub-species of the same species	support and rarity; α_i cancels	(5)
a distinct species B	energy and support; rarity omitted	(6)

Type I stands outside this scheme. It is not a contest between strengths but a fixed coin of bias d , and the loser of it forfeits nothing, because empty ground has nothing to lose.

6 Mutation: the origin of sub-species

A sub-species arises when a single cell mutates within the occupied domain of a species. The mutation changes that cell to a new sub-species type while it retains its primary species classification, and $X_i \mapsto X_i + 1$. The new sub-species is given a hue one small step from its parent's — uniformly within $\pm 7^\circ$ — and thereafter spreads by Type II competition.

The rate may be set in either of two ways, chosen by the *mutation rate* control:

- (a) a **constant rate per cell**, giving a species-wide rate mN_i ;
- (b) a **rate per cell proportional to energy**, $m\alpha_i$ per cell, giving a species-wide rate

$$R_i^{\text{mut}} = m\alpha_i N_i. \quad (8)$$

Option (b) is the default. It couples novelty to prosperity, so that rich species diversify faster and thereby hasten their own dissipation, which is the feedback the model exists to display.

Implementation. Rather than tracking a mutation clock for every cell, mutation events are drawn against the population-wide rate and then assigned. An exponential clock accumulates: each chunk subtracts $R\Delta t$ from a running debt, and whenever the debt falls to zero an event fires and a fresh $\text{Exp}(1)$ draw is added back. This is exact for a rate held constant across the chunk.

When an event fires, a cell is chosen uniformly at random from all occupied cells. Under option (a) this alone gives each species its share, since a species is selected in proportion to its area. Under option (b) the global rate is run at the envelope $m\alpha_{\max}N_{\text{occ}}$, where $\alpha_{\max}$ is the largest energy currently held by any species, and the drawn event is accepted with probability $\alpha_i/\alpha_{\max}$. This *thinning* recovers the per-cell rate $m\alpha_i$ exactly, at the cost of some rejected draws.

One consequence deserves note: if the mutating cell was the last of its sub-species, that sub-species dies as the new one is born, and X_i is unchanged. Mutation adds diversity only where there is something to mutate away from.

7 Energy

Each occupied cell supplies energy at rate E , so a species of area N_i has a total energy capacity per unit time of $N_i E$. Each extant sub-species costs β per unit time — the *cost of multiplicity*. Hence

$$\frac{d\alpha_i}{dt} = N_i E - \beta X_i. \quad (9)$$

Discretising (9) over a chunk of duration $\Delta t_K = K/N^2$,

$$\alpha_i(t + \Delta t_K) = \max\left(0, \alpha_i(t) + \Delta t_K(N_i E - \beta X_i(t))\right). \quad (10)$$

Remarks on the energy equation.

- (i) α_i does not appear on the right-hand side of (9). It is a pure accumulator — a running store of energy — not a self-regulating quantity, and it has no fixed point of its own. It grows while $X_i < N_i E/\beta$ and falls once diversity passes that threshold.
- (ii) Since $X_i \leq N_i$, the most negative attainable drift is

$$\left. \frac{d\alpha_i}{dt} \right|_{\text{extreme}} = N_i (E - \beta), \quad \text{attained at } X_i = N_i. \quad (11)$$

It follows that a species can lose energy only if $\beta > E$. This is the condition under which the catastrophe of Section 8 is reachable at all, and the interface warns when it is violated.

- (iii) A species carrying a single sub-species needs at least β/E cells to break even. Below that area its energy drains away whatever it does. At the default settings the figure is about nine cells, and it governs the fate of the smallest fragments of a burst (Section 11).
- (iv) Energy is floored at zero rather than allowed to go negative. A species at $\alpha_i = 0$ loses every Type III contest it enters against any species with energy to spend, and under the α -coupled mutation rule it stops mutating altogether. It is not dead — it still takes empty ground, and if its area exceeds $\beta X_i/E$ its energy will recover — but it is defenceless until it does.
- (v) Area helps and diversity hurts, and both linearly. A species survives by being large and internally uniform.

8 Fragmentation

When a species runs out of energy — because it has come to carry too many sub-species — it **fragments** in a catastrophe event. Every sub-species immediately breaks away and becomes a new, independent species. The domains do not move: what changes is the law by which they interact, which switches from Type II to Type III. Neighbouring patches that a moment earlier were contesting a shared inheritance now contest each other in proportion to their separate energies.

8.1 The triggers

There are two triggers. They are not alternatives: both are live at once, and they are tested in the order given.

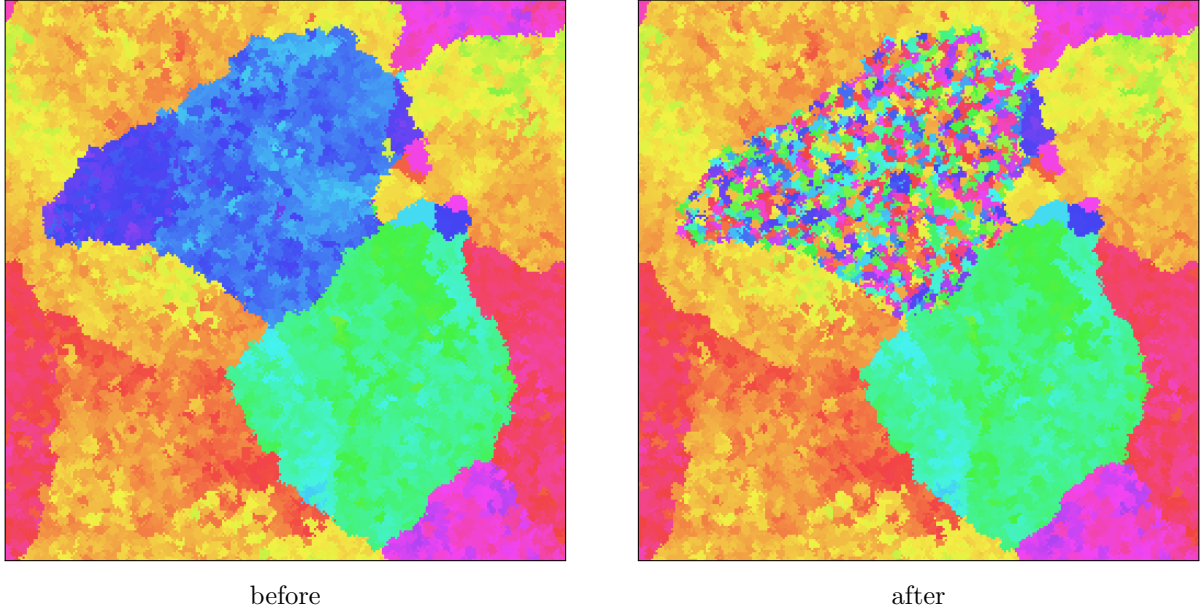


Figure 3: A burst, across the single animation frame in which it happens, at $t \approx 1239$. The species held 12,826 cells — a fifth of the lattice — and still had $\alpha = 755$ in hand when the hazard (13) fired; all 1444 of its sub-species became species at once, and it was the only species to fragment in that frame. Nothing has moved: the outline of the parent is exactly preserved, and its neighbours are untouched. What has changed is the law inside that outline, which a moment earlier was Type II and is now Type III. This panel is drawn under the redraw colour rule rather than the default, so that each fragment takes a fresh hue and the division is unmistakable; under the default the fragments would scatter about their parent’s own hue instead.

- (a) **Deterministic.** Fragment as soon as the update (10) would take the energy to zero or below:

$$\alpha_i(t) + \Delta t_K (N_i E - \beta X_i) \leq 0 \implies \text{burst.} \quad (12)$$

- (b) **Stochastic.** Otherwise fragment with probability

$$1 - e^{-R_i^{\text{frag}} \Delta t_K}, \quad R_i^{\text{frag}} = \delta \frac{X_i}{\alpha_i(t + \Delta t_K)}, \quad (13)$$

a rate increasing in diversity and decreasing in energy, evaluated at the newly updated energy.

Rule (b) diverges as $\alpha_i \rightarrow 0^+$, so it anticipates rule (a) rather than competing with it, while also permitting early bursts in a species that is merely strained. At the default settings rule (b) does essentially all the work: a species in the steady regime of Section 11 holds an energy of several hundred and never approaches zero, so it bursts because (13) eventually fires, not because it is bankrupt. Rule (a) is the floor beneath that, and matters chiefly for species too small to pay for the diversity they carry.

A species with one sub-species cannot fragment. There must be at least two sub-species for a burst to mean anything, and the trigger is suppressed below that. A monomorphic species that exhausts its energy simply sits at $\alpha_i = 0$ until its area grows enough to recover, or until it is overrun.

8.2 What the fragments inherit

Each new species takes exactly the cells its sub-species held, is given a fresh serial tag, and begins life with $X = 1$: it is its own only sub-species. Its hue is set by the colour rule of Section 9. Its energy is the one thing the burst does not settle by itself, since the parent’s store is being wound up rather than divided by any natural rule; three readings are offered, chosen by the *energy of the fragments* control:

Rule	Each fragment receives	
reset	α_0	(default)
equal split	$\alpha_{\text{parent}}/X_{\text{parent}}$	
by area	$\alpha_{\text{parent}} \cdot u /N_{\text{parent}}$	

The two splitting rules interact with the choice of trigger in a way worth noticing. A deterministic burst happens precisely when the parent’s energy has reached zero, so there is nothing left to divide and both splitting rules hand every fragment an energy of zero. Only a stochastic burst, which fires while the parent is still solvent, leaves an inheritance. The reset rule is therefore the only one that behaves identically under both triggers, which is why it is the default.

Once every sub-species has been promoted, the parent is emptied of cells and sub-species and removed from the population. The time of the burst is logged, so that it can be drawn on the graph.

9 Colour

Cells are coloured in HSB with saturation fixed at 0.72 and brightness at 0.95, and hue free, so that the entire taxonomy is carried on the hue circle: red at 0° , through orange, yellow, green, blue and indigo, to violet and back to red at 360° . Empty ground is drawn near-black.

Within a species the arrangement is fixed. The colour of a new sub-species is a small step in hue from the antecedent in which the mutation occurred — uniformly within $\pm 7^\circ$ — so hue measures genealogical distance from the founder and the species reads on screen as a band of closely related shades.

What happens *at* a burst is a matter of choice, and four rules are offered, selected by the *colour after a burst* control. Under each of them, the sub-species that is about to become a species holds the hue it has until the moment it is promoted.

Redraw. Each fragment draws a fresh hue uniformly from the circle and keeps it. The taxonomy is legible at a glance — no two fragments will be confused — but the genealogy is discarded at every burst.

Jitter. Each fragment steps off its own hue by a normal draw of standard deviation σ . The family stays recognisable; the siblings separate. This is the default, and the cost of it is that neighbouring species are often close in hue — which is what the perimeter of Section 13 is for.

Spread. Each fragment is pushed g times its own offset from the family’s mean hue, in the direction it already lay:

$$\varphi \mapsto \bar{\varphi} + g(\varphi - \bar{\varphi}), \quad (14)$$

with the difference taken as the signed separation in $(-180^\circ, 180^\circ]$ and $\bar{\varphi}$ the *circular* mean — the direction of the resultant of the unit vectors, since 350° and 10° average to 0° and not to 180° . The siblings fan apart along the lines their mutations had already started, so the branching structure survives the catastrophe that produced it.

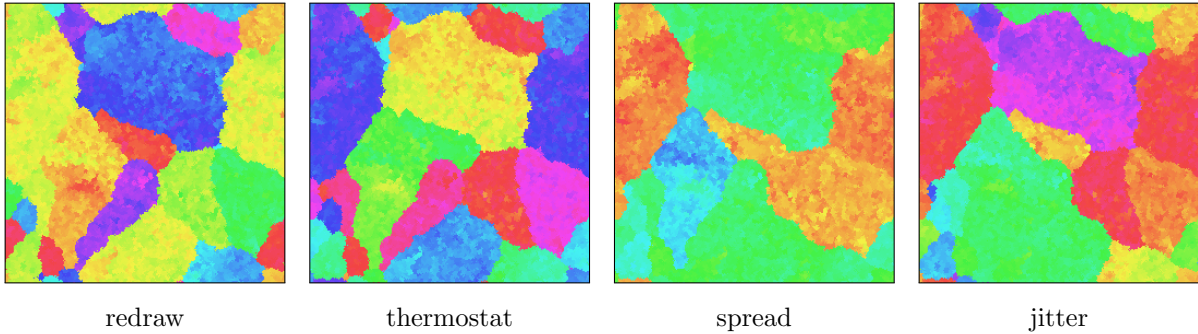


Figure 4: The four colour rules, same seed, same time ($t = 2000$). The patterns are not merely similar but identical — the lattices agree cell for cell, as the two-stream construction of Section 12 requires — so the panels differ in hue alone. Redraw scatters fragments across the whole circle. The thermostat holds each species on one of $k = 12$ settled hues, giving a quantised palette in which relatives share a colour. Spread fans siblings apart from their family mean, and jitter — the default — steps each off its own hue, so both keep some memory of descent.

Thermostat. The fragment keeps its hue and inherits, instead, a quantity of *heat* that its parent’s multiplicity paid for. It then walks away from its siblings as that heat decays. Alone among the four it makes hue a dynamical variable in its own right, obeying a stochastic differential equation on the colour circle; it is described below.

9.1 The thermostat

Under this rule each species’ hue φ obeys, in degrees and with time in sweeps,

$$d\varphi = -a \sin\left(k\varphi \frac{\pi}{180}\right) dt + \sqrt{r(t)} dW. \quad (15)$$

The two terms do different jobs.

The noise, and the heat that drives it. A fragment of a parent that carried X sub-species is given a total variance budget

$$V(X) = D^2 \frac{X}{8}, \quad (16)$$

so that its total r.m.s. displacement in hue is $D\sqrt{X/8}$ degrees. The reference multiplicity of 8 makes D the plain design dial: a species that bursts into eight fragments disperses them by D degrees, one that bursts into thirty-two disperses them by twice that. Multiplicity therefore buys colour separation in exactly the way it buys catastrophe.

That budget is spent over a settling window of T sweeps, at a variance rate falling linearly to nothing:

$$r(t) = \frac{2V}{T} \left(1 - \frac{t}{T}\right), \quad 0 \leq t \leq T, \quad (17)$$

whose integral over the window is exactly V . When the window closes the motion has eased to a stop rather than being cut off, and the heat is set to exactly zero: a settled species holds its colour precisely until it fragments in turn.

The drift, and the wells. Left to itself the noise would stop wherever it happened to be. The drift term gathers it instead. The function $-a \sin(k\varphi)$ has stable zeros at the k hues evenly spaced around the circle, at multiples of $360^\circ/k$, and pulls the walk towards whichever

is nearest with a strength set by a . A fragment does not merely stop diffusing; it is drawn into a definite colour and held there. When the settling window closes the hue is placed exactly on its well, so the palette stays crisp rather than merely approximate. Setting $a = 0$ removes the drift and leaves the walk to stop where it lies.

The band moves as a body. The walk is integrated once per species, and the resulting increment is added to the species' hue *and* to each of its sub-species' hues alike. The $\pm 7^\circ$ fine structure that mutation builds inside a species therefore survives the walk intact: the band drifts and settles as a unit, and its internal genealogy is not smeared away.

A road not taken. One could go further and dispense with discrete labels altogether, defining the model *by intervals of hue*: species identity would then be a matter of proximity on the colour circle, and speciation a matter of a lineage's hue drifting far enough from its relatives to fall outside the interval. The model does not do this, and the distinction is worth stating plainly. Even under the thermostat, hue is a consequence of the taxonomy and never a cause: it is carried along by species that the competition rules have already made, and no cell ever changes hands because of a colour.

10 The algorithm

The run is a sequence of chunks. For each chunk, with K as in Section 3:

1. **Competition.** Repeat K times: choose a cell uniformly at random, and one of its eight neighbours uniformly at random. If the labels agree, or both are empty, do nothing. Otherwise classify the pair as Type I, II or III, resolve the contest with the appropriate probability, and copy the winner's labels into the loser's cell. Update the affected N_i , and X_i if a sub-species has just been driven extinct; retire any species that has lost its last cell.
2. **Advance the clock** by $\Delta t_K = K/N^2$.
3. **Mutation.** Draw events against the population-wide rate over Δt_K . For each, pick a uniformly random occupied cell, apply the thinning test if the rate is α -coupled, and on acceptance give the cell a fresh sub-species label and a hue one small step from its parent's.
4. **Energy.** Update every α_i by (10), and record the largest for the next chunk's thinning envelope. Test each species for fragmentation by the two triggers of Section 8, collecting those that fire; then fragment them, promoting each sub-species to an independent species.
5. **Colour.** Integrate (15) over Δt_K for every species that still has heat left, subdividing the step if the drift demands it.
6. **Sample.** Whenever the clock passes a multiple of two sweeps, record a snapshot for the graph.

The visible image is repainted cell by cell as labels change, and in full once per frame whenever a burst or a hue movement has altered colours that were already on screen.

11 Structural consequences

Four consequences follow from the rules as stated, and are worth keeping in view when reading simulation output.

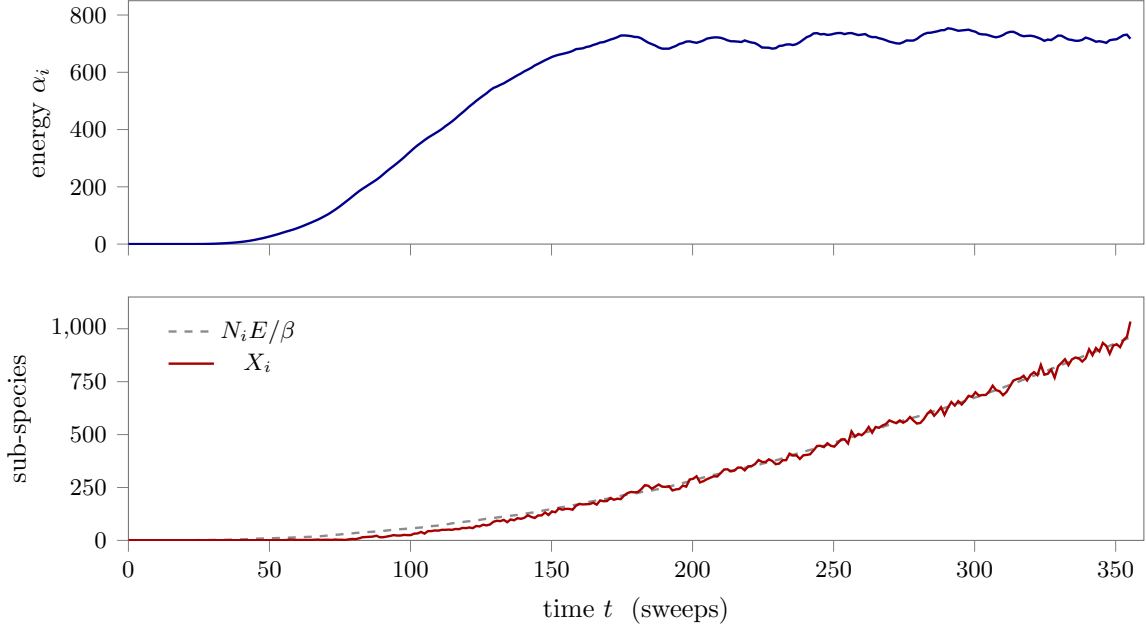


Figure 5: The founder species, followed from the first cell to its burst at $t = 356$. Its energy climbs steeply while diversity is still cheap, peaking near 754 at $t \approx 290$; thereafter mutation has brought X_i up to the balance point $N_i E / \beta$, the two curves in the lower panel lock together, and α_i goes flat. The species is not bankrupt when it bursts — it still holds $\alpha = 717$ — but by then it carries 1034 sub-species, and the hazard (13) has become large enough to fire. It is also still growing, on only 8763 cells — a seventh of the lattice. The lower panel is the whole argument of this section in one picture: the dashed line is area, rescaled, and diversity tracks it.

The occupied region never retreats. No rule ever vacates a cell. Type I cannot, by construction; Types II and III only relabel. The occupied set is therefore monotonically non-decreasing, and the lattice fills — at the default settings, by about $t = 1100$. Once it is full, Type I events cease and the dynamics reduce to relabelling under Types II and III — a biased voter model on a full lattice, punctuated by mutation and fragmentation.

The characteristic cycle. A species grows, its area drives α up, mutation accumulates sub-species, the cost βX_i overtakes the supply $N_i E$, and the species bursts into as many new species as it had sub-species. Those fragments, now mutually hostile and individually small, contest the territory their parent held. The model’s macroscopic behaviour is the statistics of this cycle repeated across a lattice. A single burst is a large event: the species of Figure 3 holds a fifth of the lattice and promotes 1444 sub-species at one stroke. At the default colonisation rate the first turn of the cycle is also complete long before the lattice is full, so growth and fragmentation overlap: the founder of Figure 5 bursts holding a seventh of the ground, with open country still ahead of it.

The population sits at the balance point. The cycle does not run away. Because mutation raises X_i while nothing lowers $N_i E$, a species is driven towards the locus

$$X_i \approx \frac{N_i E}{\beta}, \quad (18)$$

at which production and dissipation cancel and α_i goes flat. Every substantial species in a mature run is found there, to within a few per cent, with its energy parked at a few hundred

and its drift near zero. The system settles into a marginal state rather than an oscillation, and what ends a species is not bankruptcy but the accumulating hazard of (13), which grows with the diversity the balance point obliges it to carry.

Fragments are filtered by area. A burst is not a fair division. Each fragment inherits its sub-species' cells, and those areas are broadly distributed; but by remark (iii) of Section 7, a fragment of fewer than β/E cells cannot pay for even its own single sub-species. Its energy drains to zero within a fraction of a sweep, after which it loses every contest it enters and is overrun. Immediately after a large burst, hundreds of species may sit at $\alpha = 0$ simultaneously. The great majority of the species a burst creates are gone almost at once, and only the larger fragments survive to grow, accumulate diversity, and burst in their turn — which is why a run that has produced tens of thousands of species may hold only a few hundred alive.

12 Numerical and implementation notes

Random numbers, and two streams. The generator is the Park–Miller minimal standard, $x \mapsto 16807x \bmod (2^{31} - 1)$, evaluated by Schrage's trick to stay within 32-bit arithmetic. It is run as *two* independent streams, seeded from a common value: one drives the dynamics, the other drives hue noise alone. The separation is deliberate. Switching the colour rule, or the thermostat's parameters, cannot move a single cell: for a given seed the spatial trajectory is identical under all four colour rules. To keep that guarantee, the uniform hue draw used by the redraw rule is taken from the dynamics stream at every burst whether or not the rule is in force, so the stream stays aligned.

Exact integrals where they are available. Three quantities that could have been linearised are not. The mutation clock uses exponential inter-arrival times rather than a per-step Bernoulli trial. The fragmentation test uses $1 - e^{-R\Delta t}$ rather than $R\Delta t$, which matters when $R\Delta t$ is not small. And the thermostat's variance, whose rate is linear in time, is integrated in closed form over each step as $r_0\tau - \frac{1}{2}c\tau^2$, clipped at the moment the settling window closes. The total dispersal is therefore exactly $D\sqrt{X/8}$ whatever the chunk size happens to be.

Sub-stepping the colour drift. The drift in (15) stiffens as ak grows, and Euler–Maruyama is only conditionally stable. Each chunk is therefore subdivided into steps no longer than $0.24/(ak\pi/180 + 1)$, to a maximum of 64 sub-steps, before the walk is integrated. The bound is inactive when $a = 0$.

Thinning against a lagged envelope. The mutation pass runs before the energy pass within a chunk, so the envelope $\alpha_{\max}$ it thins against is the value from the previous chunk. The lag is bounded by the change in α across one chunk, which at the default settings is a fraction of a per cent of $\alpha_{\max}$.

Recycled records and stable identities. Species and sub-species records are pooled: a dead record is returned to a free list and reused, so memory tracks the high-water mark of concurrent species rather than the number ever created. Because slots are reused, every species also carries a **tag** — a serial number that is never reused — and it is the tag that labels a species on screen and keys its trace on the graph. A graph line can therefore never jump from a dead species to the live one that inherited its storage. For the same reason each sample carries the colours of the species it recorded rather than a reference to them: by the time a ghost trace is drawn its owner may be long dead and its record reissued, so the palette held in the sample is the only surviving record of what that line looked like.

Fixed constants.

Quantity	Value	Where
hue step at mutation	$\pm 7^\circ$, uniform	§6
chunk size	256 to 4096 events	§3
reference multiplicity	8 sub-species	§9
Euler–Maruyama safety factor	0.24	§12
maximum sub-steps per chunk	64	§12
graph sampling interval	2 sweeps	§13
species retained per sample	128 largest	§13
graph history	300 samples	§13
burst log	512 most recent	§13
saturation, brightness	0.72, 0.95	§9
perimeter suppressed below	2 device pixels a cell	§13

The figures. Every figure in this document was produced by the page’s own simulation core, driven headless rather than screen-captured, so the runs shown are the runs the model performs. All use seed 20260903 on a 256^2 lattice at the default settings, varying only the rule under discussion, and all are stepped in whole frames of 80,000 events — the page’s own default speed — so that the caveat of Section 13 is satisfied and the page at `?seed=20260903` follows the same trajectory. The lattice images are the model’s own pixel buffer, enlarged without interpolation, and the perimeter in Figure 6 is stroked by the page’s own code rather than a reconstruction of it.

One departure from the defaults should be declared. Figures 1 to 5 are coloured under the *redraw* rule rather than the default jitter, because redraw gives every species a hue of its own and so makes the mechanism under discussion legible; jitter keeps relatives close together, which is a virtue on screen and a nuisance in a diagram. Figure 4 shows all four rules side by side, and Figure 6 uses the defaults untouched.

13 The interface

Running. START and RESET, or the space bar and **r**. A speed slider sets the number of elementary events attempted per animation frame, from 10^3 to 5×10^5 , defaulting to 80,000 — something over seventy sweeps a second on a 256^2 lattice.

Choosing rules. Five segmented controls select the model’s switchable rules, gathered for reference in Section 14:

Control	Options	Default
Mutation rate	mN $m\alpha N$	$m\alpha N$
Within-species contests	tie $\frac{1}{2}$ occupancy λ	occupancy λ
Neighbour support, Types II and III	off on, θ	on
Colour after a burst	redraw thermostat spread jitter	redraw
Energy of the fragments	reset to α_0 equal split by area	reset

The parameters belonging to a rule appear directly beneath it and only while that rule is selected, so the panel shows what is currently in force and nothing else.

Reading the run. The graph plots one point every two sweeps for the eight largest living species, coloured as they are on the lattice, with vertical marks at every burst. Beneath those, and much fainter, run the *ghosts*: every other species in the window, ranked by the largest value it ever reached on the metric being shown and capped at the best thirty-two. A ghost that stops in mid-graph is a species that died there — by fragmenting, or by being overrun — and one that reaches the right-hand edge is alive but too small to have earned a full trace. Without them the graph would say only that a line had ceased, never whether its owner had burst or merely slipped out of the leading eight. Three quantities may be shown — area, energy α , or sub-species per species — cycled with **g**. In area mode the scale is fixed to the whole lattice and the count of empty cells is drawn as a dashed line; in the other two the ceiling auto-scales, easing downwards after a burst so the trace does not jump. Beneath, a table lists the six largest species by tag, with area, energy and X ; below that, running totals of events, occupancy, species and sub-species alive and ever created, contests by type, mutations, bursts, and the frame and event rates. A warning line appears if $\beta \leq E$, since no species can then ever lose energy and nothing will ever burst.

The species perimeter. A separate control, on by default, draws the boundary between species over the grid (Figure 6). It is a display option and not one of the rules: the layer sits on its own canvas above the lattice, reads it, and changes nothing in it.

The boundary is drawn a side at a time. Wherever two adjacent cells differ at the *species* level, each of the two species strokes that face on its own side of it, one device pixel inside its own territory, in its own colour darkened to two-fifths. A contact between species therefore carries two hairlines rather than one, and each domain is outlined in a shade of itself, so the perimeter follows the thermostat for nothing extra — the cast is taken from the colour the species is already painted in, not recomputed from its hue. Empty ground has no species and strokes nothing, so a lone species in open country draws its own outline unanswered. Faces between two sub-species of one species carry nothing at all, which is the point of the option: the hue structure inside a domain is left to speak for itself.

Because each side belongs to a different species, faces cannot be merged into runs, and the strokes are collected into one path per species and stroked in that species' colour. The layer is keyed on the event count, so it is rebuilt only on frames that actually advanced the lattice and a paused grid costs nothing. Below two device pixels to a cell the option is suppressed — a line on every face would cover the grid rather than describe it.

Typing values. Every slider's readout is editable in place. A typed value is mapped back onto the slider's own scale — logarithmic for the decade parameters, $\log_2$ for the grid size, the identity otherwise — and is then taken exactly: neither rounded to the slider's step nor clamped to its range, so a value can be set that the handle could not reach. A value outside the range leaves the handle resting at the nearer end of the track, and dragging brings it back.

Reproducing a run. The whole configuration can be given in the URL: **seed** fixes the random seed, which otherwise comes from the clock; **size** selects the lattice (64, 128, 256 or 512); **d**, **E**, **beta**, **m**, **delta**, **a0**, **lam** and **theta** set the model parameters; **disp**, **tset**, **pull**, **wells**, **mult** and **sig** set D , T , a , k , g and σ ; and **rules** takes a string of digits selecting, in order, the fragment-energy rule, the mutation rate, the within-species rule, neighbour support, and the colour rule. The defaults correspond to **rules=01113**.

One caveat on reproducing a run. The seed fixes the random streams, but not the order in which they are drawn from: the fragmentation test of Section 8 spends one draw per living species per chunk, and the chunk boundaries depend on how many events each frame resolves. Two runs from one seed therefore agree exactly only if the speed slider is left where it was. Move it and the run is still the same model, statistically, but no longer the same trajectory.

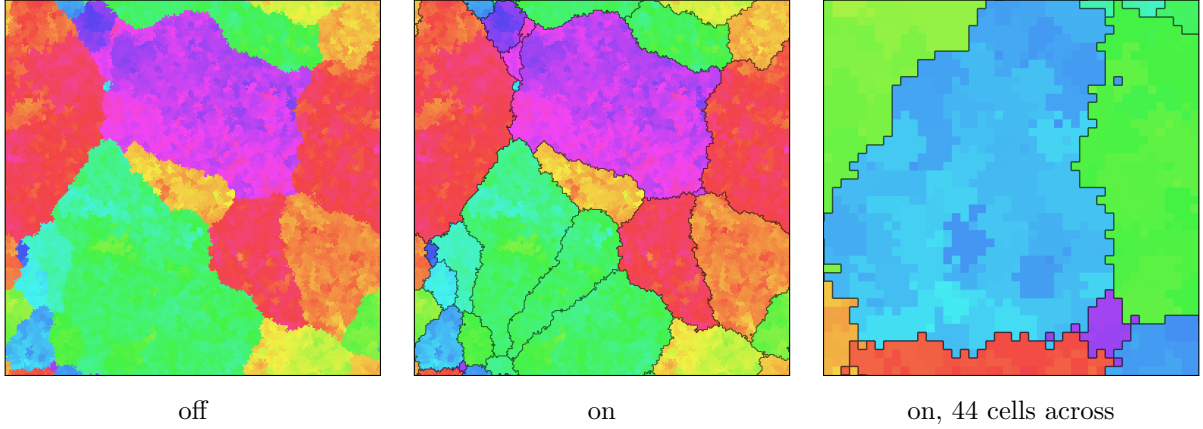


Figure 6: The species perimeter, over the same lattice at $t = 2000$, drawn at the default settings, with a magnified window at the right. Every species draws the boundary of its own territory, one device pixel inside its own cells, in a cast of its own colour darkened to two-fifths. Where two species meet you therefore see two hairlines side by side, each in its owner’s hue, and against empty ground you would see only one. The left pair shows why the option earns its place under the default colour rule: sibling species sit close on the hue circle, and several of the apparently single domains are two or three. Faces between two sub-species of a single species carry nothing, which is why the mottling inside each domain is left undisturbed: the layer reads the lattice through the same species label a Type III contest reads, and changes nothing in it.

14 Options at a glance

Five of the model’s rules are switchable rather than fixed. Each is described where it belongs; this section gathers them for reference. Every figure in this document uses these defaults except where its caption says otherwise.

Rule	Settings	Default	Section
Mutation rate	$mN_i \mid m\alpha_i N_i$	$m\alpha_i N_i$	§6
Within-species contests	tie ($\lambda = 0$) rare-type advantage λ	advantage	§5
Neighbour support	off ($\theta = 0$) on, exponent θ	on	§5
Colour after a burst	redraw thermostat spread jitter	jitter	§9
Energy of the fragments	reset to α_0 equal split by area	reset	§8

Two remarks on the first three. Neighbour support applies to Types II and III alike, while the within-species control reaches Type II only; the two are independent, and a Type II contest is a fair coin only when both are off. And the choice of fragment energy interacts with the fragmentation triggers: a deterministic burst leaves nothing to divide, so the two splitting rules then hand every fragment an energy of zero, and only the reset rule behaves identically under both triggers.

One condition is not optional. Since $X_i \leq N_i$, the steepest decline a species can suffer is $N_i(E - \beta)$, so energy can fall at all only if $\beta > E$. If $\beta \leq E$, production exceeds dissipation whatever the diversity: no species ever exhausts its store, nothing fragments, and the run reduces to the lattice filling and then relabelling indefinitely under Types II and III. The defaults satisfy $\beta > E$ by an order of magnitude, and the interface warns when the condition is violated. Beyond it the model is not delicate — the balance point of Section 11 is an attractor, not a tuning — but below it there is no model at all.

Why the less obvious mechanisms are there. Three of the rules above are not forced by the model's premise, and each earns its place by what goes wrong without it. Neighbour support gives domains surface tension; without it boundaries wander freely and the lattice dissolves into filigree (Figure 2). The rare-type advantage λ decides whether a burst is a catastrophe or merely a renaming: without it one sub-species engulfs its species, so that of the 7262 fragments the largest species would yield at $t = 2000$ exactly one is large enough to survive, while with it about half of 1534 fragments clear the threshold. Diversity in the bare sense survives either way — X_i climbs to the balance point under both settings — but only under the advantage does fragmentation actually multiply the species. And the colour thermostat makes hue dynamical and heritable, so that the picture reports genealogy rather than merely labelling it — the one mechanism of the three that changes nothing about where a cell goes, only about what it looks like when it gets there.